

Hardware Required for a Pair of HandUMI (2 Units)

Hardware needed to build 2 HandUMI units (one pair). Quantities are per pair; prices use the unit cost from the linked listing, multiplied by the quantity needed.

Mechanical & Structural

Part	Qty Needed	Unit Price	Link	Total (USD)*
PLA Filament (1.75 mm, ~200 g per unit)	0.4 kg	\$12.59 / kg	Amazon	\$5.04
Rolling Bearing (MR63ZZ, 3×6×2.5 mm)	8	\$13.29 / 20 pcs	Amazon	\$5.32
Linear Bearing (LM4UU, 4×8×12 mm)	8	\$7.49 / 4 pcs	Amazon	\$14.98
Axle (Ø4 mm × 135 mm, 304 stainless rod)	2 rods	\$11.99 / 10 pcs	Amazon	\$2.40
M3 Hardware Kit (750 pcs, screws + nuts)	10 total (2 @ M3x20mm + 4 @ M3x12mm + 4 @ M3x15mm)	\$9.99 / 750 pcs	Amazon	\$0.13
Hook-and-loop Strap (6 in / 15 cm, reusable)	8	\$13.99 / 75 pcs	Walmart	\$1.49
M2 Hardware Kit (660 pcs, screws + nuts)	32 total (16 screws + 16 nuts)	\$7.99 / 660 pcs	Amazon	\$0.39
Subtotal				\$29.75

Electronics

Part	Qty Needed	Unit Price	Link	Total (USD)*
USB Camera (SVPRO IMX335, 5 MP, wide-angle UVC)	2	\$50.69	AliExpress	\$101.38
Feetech Servomotor (STS3215, 7.4 V)	2	\$13.89	Alibaba	\$27.78
Power Supply (5 V, 5.5×2.1 mm barrel)	2	\$10.00	Amazon	\$20.00
Servo Controller (Waveshare Bus Servo Adapter (A), USB/UART)	2	\$10.60	Amazon	\$21.20
USB-C Cable (USB-A to USB-C, braided)	2	\$7.00 / 2 pcs	Amazon	\$7.00
Subtotal				\$177.36

GRAND TOTAL for a pair (2 units, approx.)	\$207.11
Per unit (approx.)	\$103.56

***Unit Price** reflects the real per-unit cost from the linked listing (e.g. pack price ÷ pack quantity); Total = Unit Price × Qty Needed. Prices are early reference estimates and may vary with shipping, region, and availability.

Note: this BOM does not include the VR headset (e.g. PICO 4 Ultra / Meta Quest) used for the shared tracking layer. That is a separate, one-time purchase, not duplicated per HandUMI unit.